

Movie Theater Seat Reservation System

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Reel Solutions continued development of the Movie Theater Seat Reservation System by implementing core catalog, ticketing, seat, and reservation components. The team performed unit testing, resolved formatting and validation issues, and updated the sprint board to reflect progress toward next week's demonstration.

Current Team Progress

Reel Solutions has progressed from requirements gathering and architectural design into Java implementation and testing for the Movie Theater Seat Reservation System. In Week 1, the team defined the project vision, preliminary requirements, user stories, acceptance criteria, minimum viable product scope, and a four-week sprint plan. In Week 2, the team completed the Customer and Theater Manager Use Case Views, Logical View, and Process View. These diagrams define the system's main actors, major classes, relationships, and seat-reservation workflow.

This week, the developers implemented several classes identified in the architectural design. The current Java project includes the Movie, Showtime, Seat, Reservation, Ticket, BookingConfirmation, Main, and ReservationTest classes. Together, these classes handle movie and showtime information, ticket pricing, seat accessibility and availability, reservation creation, duplicate reservation prevention, cancellation, and booking confirmation.

The provided Java project compiles and runs successfully. The current test drivers verified ticket-price formatting, readable date and time formatting, booking confirmation generation, rejection of missing reservation information, wheelchair-accessible seating, successful seat reservation, duplicate-reservation prevention, cancellation, and seat release after cancellation.

Requirements and User Stories Addressed This Week

Reel Solutions prioritized the user stories tied to the system's core reservation workflow. They selected these stories because they represent the minimum functionality needed to demonstrate a working reservation system. The development completed this week supports the following requirements and user stories:

- Customers can view movie and showtime information.
- Customers can identify whether a seat is available.
- Customers can identify wheelchair-accessible seating.
- Customers can select and reserve an available seat.
- The system prevents reserving an already reserved seat.
- The system validates required reservation information.
- Customers can view properly formatted ticket pricing.
- Customers can receive a booking confirmation.
- Customers can cancel an active reservation.

- A canceled seat becomes available for another reservation.

The work also supports theater-management requirements. The Movie and Showtime classes provide a foundation for managing movie titles, screening times, ticket prices, and sold-out status. Accurate seat and reservation information will also let theater managers monitor reserved and available seats and track expected attendance as the remaining components integrate.

Development and Unit-Testing Work

Development for this week was primarily on building the core catalog and ticketing components. This included the Movie, Showtime, Ticket, and BookingConfirmation classes and a Main driver class. Unit tests were executed properly within the Main class to verify accurate ticket formatting, successful booking confirmation generation, and system resilience against invalid data.

Issues and Unexpected Behaviors

During testing, a few formatting and data-handling issues were noticed that disrupted the workflow. To start, the default output for the LocalDateTime objects that were generated appeared too clunky which would make the booking confirmation hard to read for a regular customer. The second thing was managing ticket prices using standard decimal variables presented a high risk of precision errors. Lastly, testing had revealed that attempting to generate a confirmation with missing or null reservation data would cause the system to crash.

Addressing the Issues

To strengthen the reliability of the reservation workflow and improve the clarity of system outputs, Reel Solutions reviewed each issue identified during testing and applied targeted refinements. These updates focused on improving data formatting, preventing avoidable failures, and ensuring that reservation information is consistently validated before processing.

- Integrated Java's DateTimeFormatter in both the Main and BookingConfirmation classes to convert the raw timestamps into clean, readable dates
- Refactored the Showtime and Ticket classes to use Java's BigDecimal class, which helped with precise calculations and enforcing a strict two-decimal format for all prices
- Added a validation gate in the BookingConfirmation constructor. This helped catch and reject empty reservation data which prevented the system from crashing

Seat and Reservation Development Work

This week's development work consisted of developing the seat and reservation functionality for the Movie Theater Seat Reservation System. The Seat class was developed to hold the row, seat number, wheelchair accessibility, and reservation status. The class

also includes methods to determine if a seat is available, reserve the seat, release the seat in the event of a cancellation, and return the seat label.

A Reservation class was also developed to connect the customer, chosen showtime, and seat. Before a reservation is made, the system verifies that the customer email, showtime, and seat are valid. The system also ensures that the seat the customer wants is available. Once the seat is reserved, the system prevents any other reservations for that specific seat from being made.

Reservation Workflow

This implementation follows the reservation workflow that was developed in Week 2. This functionality consists of checking seat availability, identifying whether wheelchair-accessible seating is available, selecting an available seat, making a reservation, avoiding duplicate reservations, and cancelling a reservation if one has already been made.

Testing Done

The testing was conducted using ReservationTest.java, which tested the functionality of reserving a seat, attempting to duplicate a reservation for an already reserved seat, cancelling a reservation, seat availability, and wheelchair accessibility. The first reservation was completed successfully, and the second reservation for the same seat was correctly blocked by the application. The cancellation test also verified that the seat became available again after cancelling the reservation. The wheelchair accessibility functionality also worked as expected.

Problems and Verification

No significant problems were found during testing. One crucial test case was attempting to reserve a seat that was already booked. The application correctly blocked the duplicate reservation and showed an error message. The reservation logic also uses a verification process to prevent a reservation from being made when required information is missing.

Integration and Future Plans

The Seat and Reservation classes were tested with the previously developed classes and did not show any errors. Prior to next week's presentation, the reservation process will continue to be reviewed, and additional testing can be completed to ensure that the system works properly.

Alignment With the Project Goals and Architecture

The Java implementation remains consistent with the project vision and the architectural design completed during Week 2. The process begins with a movie and showtime, continues through seat selection and availability validation, and results in a reservation, ticket information, and booking confirmation. The implemented classes closely follow the major objects and relationships presented in the Logical View.

Reel Solutions will continue comparing each implementation task against the original user stories and acceptance criteria. A feature will be considered complete when it supports the intended user need, handles invalid information appropriately, compiles successfully, and passes the relevant tests. This approach helps prevent the team from adding functionality outside the approved project scope.

Decisions and Priorities

Reel Solutions has maintained the decision to implement a small but functional portion of the minimum viable product instead of attempting to complete every proposed feature. The highest priority is completing and integrating the reservation workflow before next week's presentation. The team's immediate priorities are:

- Merge the catalog, ticketing, seat, reservation, and confirmation components into one connected workflow.
- Connect BookingConfirmation directly to the completed Reservation information.
- Replace the hardcoded demonstration data in Main.java with a console-based interface using Java's Scanner class.
- Continue testing missing data, unavailable seats, duplicate reservations, cancellation, pricing, and booking-confirmation behavior.
- Update the sprint board to reflect completed, in-progress, and remaining work.
- Prepare a stable demonstration of the reservation workflow for next week.

Reel Solutions also decided to use `DateTimeFormatter` for readable dates and times and `BigDecimal` for accurate monetary values. The team added validation to prevent missing reservation data from causing a system failure. These refinements improve the application's reliability while keeping development aligned with the original requirements.

Team Coordination and User Support

The Product Owner will continue reviewing the implementation against the project vision, user stories, and acceptance criteria. The Scrum Master will maintain the sprint board and monitor task movement, while the developers complete integration, testing, debugging, and presentation preparation. Team check-ins and GitHub updates will identify blockers and ensure parallel development work integrates correctly.

The current work directly supports customer needs by providing accurate seat availability, accessible-seat identification, reservation validation, pricing, confirmation, and cancellation. It also supports theater-manager needs by maintaining reliable movie, showtime, pricing, sold-out, seat-status, and reservation information. Keeping accurate shared data for both groups remains a central priority as the team completes integration and prepares the system demonstration.

Sprint Board Update

Reel Solutions updated the sprint board this week to reflect the team's development progress and the status of ongoing tasks. Several core reservation workflow items, including seat selection, duplicate-reservation prevention, ticket formatting, and booking confirmation, have moved into the "Done" column. Integration tasks and the development of the console-based interface remain in "In Progress," while upcoming theater-management features are still in "To Do" for next week's sprint. The sprint board continues to serve as a clear record of completed work and remaining priorities as the team prepares for the next phase of development.

Repository: <https://github.com/TatianaH25/Movie-Theater-Seat-Reservation-System>

Project Board: <https://github.com/users/TatianaH25/projects/1>